

The Virality Paradox: Engineering Neural Synchrony to Create a Track from Raw Brainwave Data

From Brainwaves to Sound: Methodologies for Neural Sonification

The ambition to compose music directly from the electrical activity of the brain represents a convergence of neuroscience, computer science, and artistic expression. While the notion may seem futuristic, its foundations are built upon decades of research in data sonification—the process of converting data into audible sound [114](#). In the context of neuroscience, this practice has evolved from rudimentary experiments to sophisticated systems capable of translating complex brain signals into structured auditory experiences. The core of this endeavor lies in Electroencephalography (EEG), a non-invasive technique that measures voltage fluctuations resulting from ionic current flows within the neurons of the brain [24](#). As one of the standard methods for measuring brain activity and the primary source of signals for noninvasive Brain-Computer Interfaces (BCIs), EEG provides the necessary raw material for any attempt at neural-to-musical translation [24](#) [70](#). However, the path from raw EEG data to a coherent musical note is fraught with technical challenges and requires a deliberate framework for interpretation.

The history of converting brain signals into sound begins not in a laboratory but in the realm of experimental music. The seminal work in this area is Alvin Lucier's *Music for Solo Performer* (1965), often referred to as the 'brain wave piece' [46](#) [103](#). In this composition, Lucier attached an EEG electrode to his scalp, which was connected to a drum set. His alpha brainwaves, when relaxed, would trigger a transistor amplifier to activate the drumsticks, causing them to strike the instrument [16](#) [104](#). This piece is a landmark because it uses the brain as a reactive performance instrument; the music is a direct, albeit indirect, consequence of the performer's state of consciousness. It is not a composition written by the brain, but a sound generated by it. This work established the foundational idea of using neurophysiological signals as a source for art, but its methodology is fundamentally different from creating a structured musical track. It prioritizes the concept of a brain-driven performance over the aesthetic qualities of the resulting sound.

Contemporary research has moved beyond this purely reactive model toward systematic sonification, where specific features of the EEG signal are mapped to specific sonic parameters. The goal shifts from mere representation to creating an auditory display that can convey information about cognitive states or, more ambitiously, to generate aesthetically pleasing music [102](#). The Oxford Academic defines sonification as the mapping of data dimensions to sound parameters, which can be physical (frequency, amplitude), psychophysical (pitch, loudness), or perceptual [9](#). This mapping process is the crux of neural composition. The four essential parameters of a musical note—pitch, volume, duration, and timbre—serve as the target domain for this translation [7](#).

Pitch is the most frequently mapped parameter. One straightforward method involves translating the amplitude or power of an EEG signal directly into a change in pitch [2](#) [113](#). For instance, a study by Lu et al. demonstrated a method to generate scale-free brainwave music by translating the amplitude of EEG signals to musical pitch according to a power law, a mathematical relationship found in many natural phenomena [2](#) [84](#). Another approach assigns different EEG frequency bands (Delta, Theta, Alpha, Beta, Gamma) to different musical scales or tonal centers, musically representing distinct mental states associated with those frequencies [28](#). Volume, or loudness, is commonly linked to the energy or amplitude of the EEG signal, providing dynamic contrast to the piece [7](#) [113](#). Duration can be mapped by correlating the period of an EEG waveform to the length of a musical note, creating a direct link between the speed of neural oscillations and the track's rhythmic structure [80](#) [94](#). Timbre, the quality or color of a sound, is less commonly addressed but could be modulated based on the spectral content of the EEG signal or by blending multiple sound sources triggered by different neural frequency bands.

A critical challenge in this entire process is signal integrity. EEG signals recorded from the scalp are notoriously susceptible to contamination from various sources, including muscle activity (electromyogram or EMG), eye movements (electrooculogram or EOG), and even hair movement [16](#) [69](#). These artifacts can easily mask the underlying neural signals, leading to distorted mappings and nonsensical musical output [23](#). Therefore, robust preprocessing and noise reduction techniques are paramount for any real-time BCI system [69](#). Researchers have developed custom drivers and optimized system performance to improve signal quality [48](#), and studies have investigated how the placement of the EEG headset can significantly impact data accuracy [56](#). Open-source toolboxes, such as BioPyC, provide platforms for offline processing and classification of biosignals, offering a starting point for developing the necessary pipelines to clean and prepare the raw neural data for sonification [60](#).

While EEG is the most common modality due to its portability and temporal resolution, other neuroimaging techniques offer complementary insights. Functional Magnetic Resonance Imaging (fMRI) provides superior spatial resolution, allowing researchers to pinpoint activity in specific brain regions [22](#) . Recent studies have explored using simultaneously recorded fMRI and EEG data to produce "EEG-fMRI music," representing a step toward more comprehensive brain music [74](#) . Furthermore, deep neural networks have been trained to reconstruct natural sounds from fMRI data, suggesting the potential for decoding complex auditory information directly from brain activity [85](#) . However, the immobility and high cost of fMRI make it impractical for interactive music-making. A more promising alternative is functional Near-Infrared Spectroscopy (fNIRS), which measures blood flow changes and is less susceptible to motion artifacts than fMRI, though it also has lower temporal resolution [23](#) [105](#) . The combination of fNIRS and EEG, known as multimodal fusion, is an active area of research aimed at leveraging the strengths of both modalities [23](#) . For the immediate goal of creating a real-time, interactive musical track, EEG remains the most viable technology. It is important to acknowledge, however, the significant gap between what EEG measures—gross electrical activity from populations of cortical neurons—and the user's interest in "synaptic activity." True translation of individual synaptic firings or spike trains requires invasive recording methods, such as implanted Utah Array electrodes, which are currently limited to medical applications in tetraplegic patients [106](#) . Thus, the focus must remain on using EEG as a proxy for broader cognitive and emotional states.

Parameter	Mapped Neural Feature	Mapping Strategy Example	Relevant Study/ Concept
Pitch	EEG Amplitude / Power	Direct or logarithmic mapping to musical pitch.	2 113
Pitch	EEG Frequency Band	Assigning Delta, Theta, Alpha, Beta, Gamma bands to different musical scales.	28
Volume	EEG Energy / Amplitude	Mapping signal energy to the loudness of a note.	7 113
Duration	EEG Waveform Period	Directly mapping the period of a waveform to the length of a note.	80 94
Timbre	EEG Spectral Content	Modulating sound texture based on the frequency spectrum of the signal.	7

In summary, the direct translation of brainwaves into sound is a scientifically plausible and well-documented field of inquiry. It has evolved from the conceptual art of Lucier to a more systematic practice of sonification, grounded in the measurable properties of EEG signals. By mapping features like amplitude, power, and frequency to fundamental musical parameters such as pitch, volume, and duration, it is possible to generate an initial stream of sound from thought. However, this raw sonification often results in

atonal and chaotic noise rather than a structured musical composition. Achieving coherence requires moving beyond simple one-to-one mappings and developing more sophisticated architectural frameworks that impose musical logic onto the generated data.

Beyond Data Representation: Architecting Coherent Musical Structures

The transition from raw neural sonification to a compelling musical composition hinges on solving a central paradox: how to harness the unique, personal, and often unpredictable nature of brain signals while imposing the structural constraints necessary for musicality. Simple parameter mapping, while a valid first step, typically produces dissonant and rhythmically erratic audio that lacks the melodic hooks, harmonic progressions, and formal architecture that define popular music [102](#). To create a track that is not just "organically derived from cognitive processes" but is also genuinely listenable and emotionally resonant, a more advanced framework is required—one that acts as an intelligent intermediary between the neural input and the final sonic output. This involves integrating rule-based systems, machine learning, and biologically inspired computing models to sculpt the raw data into a coherent artistic form.

Early efforts in EEG-to-music translation laid the groundwork for such frameworks. For example, D. Wu's 2010 paper proposed a method to translate human EEG into music to represent mental states, establishing foundational rules for the process [8](#) [29](#). Similarly, a study by Lu et al. in 2009 detailed translation rules that included mapping the period of an EEG waveform to note duration and the change in average amplitude to note volume [94](#) [111](#). While these rule-based approaches were a significant step forward, they still treat the brain as a direct input device without a higher-level understanding of music theory. A truly coherent composition requires a system that understands concepts like scales, keys, chords, and rhythmic patterns. An enhanced rule-based system could, for instance, map the amplitude of an EEG signal not to any arbitrary pitch, but to the nearest available note within a predefined chord progression or a specific mode (e.g., major for a positive emotional state, minor for a negative one). This transforms the process from simple data-to-sound conversion into a guided compositional act.

This is where machine learning (ML) and artificial intelligence (AI) become indispensable, even within the user's goal of moving "beyond algorithmic generation."

The term "algorithmic generation" often carries a pejorative connotation of soulless, formulaic AI music. However, ML algorithms are powerful tools for feature extraction, classification, and pattern recognition, which are essential for making sense of complex neural data and guiding a creative process. For example, supervised deep learning frameworks can be trained to recognize emotional states from EEG data with high accuracy, sometimes exceeding 97% in classifying emotions into positive, negative, or neutral categories [38](#) [42](#) [44](#). These emotion classifiers can serve as high-level controllers in a composition engine. If the system detects a shift from a "neutral" to a "positive" emotional state in the user's brain, it could trigger a change in the music's key from minor to major, increase the tempo, or introduce brighter timbres. This creates a feedback loop where the music dynamically reflects the user's internal emotional landscape, adding a layer of intentional expression.

Furthermore, generative models, particularly those based on deep learning, offer pathways to create more complex and musically plausible sequences. Research has shown that Transformer models can be used to reverse-map relationships from EEG data to music, effectively learning the intricate connections between neural activity and musical output [95](#). Other studies have utilized sequence-to-sequence Long Short-Term Memory (LSTM) networks to generate music based on emotional EEG inputs [11](#). A deep learning framework has also been proposed to generate MIDI sequences that are aligned with labeled emotion predictions from EEG features, demonstrating a direct pipeline from neural signal to standardized musical notation [36](#). These models can learn the statistical regularities of musical language and use them to fill in or predict plausible musical continuations based on the initial neural prompts, thereby overcoming the inherent randomness of simple parameter mapping.

Perhaps the most biologically plausible computational paradigm for this task is the Spiking Neural Network (SNN). Unlike traditional artificial neural networks that process information in discrete steps, SNNs are designed to mimic the brain's own information processing style, where data is encoded in the timing of action potentials or "spikes" [68](#) [90](#). SNNs are highly efficient in terms of energy consumption and excel at processing temporal information, making them an ideal candidate for analyzing the time-series nature of EEG and neural spike train data [25](#) [90](#). Research has already explored using SNNs for tasks like acoustic modeling in large vocabulary recognition and for modeling brain-inspired mechanisms for musical information processing [134](#)[140](#). An SNN-based BCI could potentially translate synaptic-level timing information into sonic events in a way that feels more "organic" and less computationally mediated than traditional digital signal processing. For instance, the fluctuation in oscillation frequency that controls spike timing could be directly mapped to subtle variations in pitch or rhythm, creating a more

nuanced and lifelike musical expression [82](#). Other research has focused on using Convolutional Neural Networks to estimate synaptic connectivity from spike trains, hinting at the potential to decode more complex network dynamics [135](#).

A practical implementation of such a system would involve a multi-stage pipeline. First, high-quality EEG data would be acquired using a headset optimized for low artifact interference [48](#) [56](#). Second, the raw signal would undergo rigorous preprocessing to filter out noise and isolate relevant frequency bands [69](#). Third, a feature extraction module would analyze the cleaned signal to identify salient characteristics, such as band power, phase synchronization across brain regions, or event-related potentials [26](#) [93](#). Fourth, a hybrid mapping engine would apply the composition rules. This engine could combine simple, direct mappings (e.g., Alpha band power -> overall volume) with outputs from ML classifiers (e.g., detected emotion -> musical mode). Finally, a structural layer would organize the resulting stream of notes into a cohesive whole, perhaps by applying rules of counterpoint, harmony, or formal design (e.g., verse-chorus-bridge structure). This final layer ensures the piece is not just a collection of interesting moments but a complete artistic statement. Tools like Max/MSP-Jitter and Pure Data are often used for implementing such synthesis and control systems, providing a flexible environment for real-time audiovisual output [63](#). The development of a novel OpenBCI framework further demonstrates the trend toward integrated, open-source solutions for EEG-based biofeedback and sonification [48](#).

In conclusion, achieving musical coherence from neural data requires moving far beyond simple, direct translations. It necessitates the construction of a sophisticated, multi-layered system that acts as a bridge between the unstructured output of the brain and the structured world of music. By combining foundational rule-based systems with the pattern-recognition power of machine learning and the biological plausibility of spiking neural networks, it is possible to architect a framework that translates raw thought into a structured, expressive, and ultimately compelling musical composition. This approach respects the user's desire for an organic derivation from cognitive processes while acknowledging that true artistry often emerges from a dialogue between raw inspiration and deliberate structure.

The Virality Paradox: Bridging Personal Neurology and Public Taste

The ultimate objective of composing a "viral track" from brainwaves introduces a profound and challenging paradox. On one hand, the project is defined by its pursuit of authenticity and originality, seeking to create music that is organically derived from the unique, private neural signature of a single individual—a "pure brain power" origin story [16](#). On the other hand, music virality in the contemporary cultural landscape is not a function of pure novelty or technical innovation alone; it is a complex phenomenon driven by factors that resonate deeply with mass audiences [19](#). A track born from the idiosyncrasies of one person's thoughts risks being too structurally dissonant, emotionally alien, or simply incomprehensible to become widely popular. The central question, therefore, is how to reconcile the deeply personal genesis of the music with the public-facing requirements of cultural resonance and shareability.

To address this, we must first deconstruct the concept of "virality." It is not a monolithic trait but a multifaceted outcome of interactions between intrinsic song properties, promotional strategies, and the dynamics of social media platforms [19](#). Key metrics of success include streaming volumes, engagement rates (likes, shares, comments), and the duration of a song's viral lifespan [137138](#). Platform dynamics play a crucial role; short-form video platforms like TikTok, for example, are potent engines of modern music discovery, with users on these platforms being significantly more likely to engage with songs they encounter there [17](#) [19](#). The temporal evolution of a viral hit can also be complex, with songs sometimes experiencing rapid ascents followed by quick declines, or, more intriguingly, enduring popularity that resurges months after release [18](#) [120](#). Understanding these external drivers is essential, but the most critical insight for this research comes from the field of neuroscience itself, which offers a direct window into the biological underpinnings of musical appreciation.

A groundbreaking line of research has identified a powerful predictor of public musical taste: neural synchrony. In a series of studies, researchers found that the degree to which listeners' brains respond in a similar, coordinated manner to a piece of music is a significant predictor of its public appreciation on platforms like Spotify [49](#) [50](#) [51](#). This effect was observed not just three weeks after an album's release, but also ten months later, suggesting that neural synchrony captures a durable quality of the music that correlates with long-term popularity [49](#) [120](#). This finding is of paramount importance because it provides a concrete, measurable, and neurologically-grounded target for a composition system. Instead of guessing what might be appealing, a neuro-adaptive

system could be designed to optimize its output specifically to induce this synchronized response in a population of listeners.

Further research illuminates the mechanisms behind this phenomenon. Music listening is not a passive act; it engages a complex interdependence of attention, emotion, and action ⁷⁷. The brain's response to music is deeply tied to emotional regulation, with listeners actively choosing music to align with or shift their desired emotional states ⁴⁵. Specific neural markers, such as frontal EEG coherence, have been linked to verbal learning and memory, suggesting that engaging music may leave a stronger neural imprint ¹¹⁰. The asymmetry in EEG signals, for instance, can indicate a subject's emotional valence when listening to music ⁴³. This body of evidence collectively paints a picture of a listener who is an active participant, whose brain is being shaped and engaged by the sonic structures they hear. A piece of music that successfully guides this engagement, evokes a shared emotional experience, and fosters a sense of cognitive and affective unity among its listeners is, according to the data, more likely to succeed.

This leads us back to the central paradox. A track composed entirely from the raw, unfiltered neural data of one person is statistically unlikely to produce the kind of widespread neural synchrony that drives virality. Its unique phrasing, atypical harmonic choices, and unconventional rhythms may be fascinating from a personal perspective but could fail to connect with the ingrained cognitive and emotional expectations of a general audience. The solution, therefore, is not to abandon the goal of creating a neurally-derived track, but to refine the creative process. The answer lies in designing a neuro-adaptive system—a closed-loop BCI—that uses the composer's brain as a unique creative catalyst but employs principles of music cognition and neuroscience to guide the final output toward greater public resonance.

Such a system would operate on a feedback loop. During the composition process, the system could generate a preliminary version of the track based on the user's neural data. This track would then be played for a small panel of test listeners while their own EEG is recorded. The system would analyze the neural data from these listeners to measure the level of neural synchrony it induces. If the synchrony is low, indicating a lack of shared engagement, the system could automatically or semi-autonomously adjust its own internal parameters. It might nudge the generated melodies to follow more familiar contour patterns, simplify complex rhythms, or shift harmonies toward more consonant intervals. This process, known as biofeedback, allows the system to iteratively refine the composition in real-time based on direct physiological feedback about its effectiveness ⁵⁵ ⁶⁵. The result would be a hybrid artifact: a piece whose raw material is uniquely personal and "organic," but whose final form has been sculpted through neuro-feedback

to maximize its potential for inducing the very brain responses that correlate with widespread public appreciation.

This approach reframes the challenge from one of pure translation to one of intelligent adaptation. It embraces the user's desire for a direct link to thought while pragmatically acknowledging the realities of music perception and cultural transmission. The "next viral synapse" is not a single point of origin but a dynamic interface between an individual's mind and the collective neural response of an audience. By building a system that learns to navigate this interface, it becomes possible to create a track that is simultaneously a genuine reflection of a singular cognitive process and a powerful vessel for shared human experience. The final product would carry the implicit promise of the research goal: a song that feels as if it came from the very essence of thought, yet resonates so deeply with others that it finds its way into the cultural bloodstream.

Synthesizing the Vision: A Hybrid Framework for Neuro-Adaptive Composition

The investigation into composing a viral music track from raw brainwave data reveals a path forward that is neither purely deterministic nor entirely stochastic. It requires a sophisticated hybrid framework that synthesizes the user's vision for "pure brain power" with the empirical realities of musical structure and public reception. The feasibility of this endeavor rests on moving beyond simplistic, one-to-one data sonification towards a neuro-adaptive system. Such a system would leverage the unique neural signature of a creator as a primary input source while employing principles of neuroscience and machine learning to sculpt the output into a coherent and culturally resonant composition. This concluding section outlines the core components of this proposed framework and presents a phased research plan to realize this ambitious goal.

The proposed framework is built on three pillars: direct neural input, intelligent processing, and neuro-feedback optimization. The first pillar is the acquisition of high-fidelity EEG data. Utilizing modern, consumer-grade or research-grade EEG headsets that are optimized for minimizing motion artifacts is the foundational step [48](#) [56](#). The second pillar involves a multi-stage computational pipeline. This pipeline begins with rigorous signal preprocessing to remove noise from muscle and ocular movements, ensuring the integrity of the neural signal [69](#). Following preprocessing, the system extracts meaningful features from the EEG data. This goes beyond simple amplitude and includes calculating power in specific frequency bands (Alpha, Beta, etc.), measuring inter-hemispheric

asymmetry as an indicator of emotional valence, and identifying microstates that correspond to transient cognitive events [26](#) [43](#) [57](#). These features are then fed into a third stage: a hybrid mapping engine. This engine combines two types of logic. First, a set of foundational, rule-based mappings translates basic neural metrics to musical parameters—for example, Alpha band power might be mapped to the overall volume of the piece, and the period of a dominant oscillation could determine the beat subdivision [94](#) [113](#). Second, this rule-based layer is augmented by a machine learning component. Emotion classifiers, trained on datasets that correlate EEG patterns with emotional states, can act as high-level switches, altering the musical mode (major/minor), tempo, or instrumentation based on the user's inferred emotional state [36](#) [38](#). This creates a dynamic, responsive composition that is more than a static translation of data.

The third and most innovative pillar is the neuro-feedback optimization loop. This addresses the central challenge of achieving virality. The system would not produce a final track in isolation. Instead, it would present its evolving composition to a small group of test listeners while their own EEG is recorded. The system would then compute a metric of neural synchrony between the listeners' brain responses [49](#) [50](#). This metric serves as a quantitative measure of the track's ability to engage and unify its audience on a neural level. Low synchrony would trigger an adaptive response within the system. Drawing from a library of musical transformations informed by music cognition research, the system could subtly alter the composition—for instance, simplifying a complex rhythmic motif, reinforcing a harmonic cadence, or introducing a recurring melodic phrase to enhance memorability. This process of generating, testing, and refining based on real-time neurophysiological feedback continues until the composition achieves a target level of neural synchrony. The final product is thus a dual-origin artifact: its genesis is the unique neural activity of the creator, but its form has been optimized by the collective neural response of its intended audience. This framework directly confronts the "virality paradox," transforming it from a contradiction into a solvable engineering problem.

To operationalize this vision, a phased research program is proposed:

Phase 1: Baseline Sonification and Coherence Analysis. The initial goal is to establish a proof-of-concept system. Using an open-source EEG platform like OpenBCI and a toolbox like BioPyC for signal processing, a basic mapping will be created, such as translating the amplitude of the Alpha wave to the pitch of a single-note drone [48](#) [60](#). The resulting audio will be analyzed for its structural properties—does it contain discernible rhythmic patterns, dynamic contours, or melodic fragments? This phase focuses on

answering the fundamental question of whether any recognizable musicality can emerge from raw neural data before any sophisticated structuring is applied.

Phase 2: Structural Enhancement with Rule-Based Constraints. Building on the baseline, this phase introduces a more sophisticated rule-based compositional layer. The system will be programmed to map neural data onto a predefined musical structure, such as a repeating 12-bar blues progression or a simple diatonic scale. For example, the instantaneous power of the Beta band could determine which chord is played, while the relative power of the Theta and Gamma bands could modulate the timbre of the instrument. The output from this phase will be evaluated for its adherence to musical conventions and its perceived aesthetic quality through qualitative assessment by musicians and composers.

Phase 3: Implementing the Neuro-Feedback Loop. This is the core of the neuro-adaptive approach. A small cohort of participants (e.g., 10-15 individuals) will be recruited. Each participant will wear an EEG headset while listening to the compositions generated by the system during Phase 2. Their neural data will be collected and processed to calculate a real-time neural synchrony score. The system will then use this score as a fitness function to drive an evolutionary or gradient-descent algorithm that modifies the composition's parameters (e.g., chord voicing, rhythmic density, melodic intervallic content). This iterative refinement process will continue, producing a new version of the track with each cycle of feedback. The goal is to observe whether the system can learn to generate compositions that elicit higher levels of neural synchrony over time.

Phase 4: External Validation and Virality Assessment. The final, optimized track produced by the system will be subjected to a larger-scale test. It will be released on social media platforms like TikTok and Spotify, and its performance will be tracked using standard metrics for virality: streaming numbers, shares, playlist inclusions, and user-generated content [137138](#). This phase provides the ultimate test of the framework's efficacy. Did the neuro-adaptive process successfully translate the unique neural signature of the creator into a track with the potential for mainstream resonance? The results will provide invaluable data on the relationship between creator neurology, audience neural synchrony, and commercial success.

In synthesis, the creation of a viral track from brainwaves is not only conceivable but points toward a new paradigm in music composition. It moves beyond the limitations of either purely automated AI generation or purely reactive artistic performance. By embracing a hybrid, neuro-adaptive framework, it becomes possible to create music that is both profoundly personal—a direct echo of thought—and universally appealing,

engineered to resonate with the shared neural architecture of a global audience. This research charts a course from speculative possibility to tangible, scientifically-grounded practice, redefining the boundaries of creativity in the age of brain-computer interfaces.

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